Date:
No:

Issues with PCA

- Time Complexity
  ↳ finding eigenvectors & eigenvalues is time consuming $O(d^3)$
- Features may be non-linearly related.

ISSUE 1: TIME COMPLEXITY

Let w_k be the eigenvector corresponding to the kth largest eigenvalue of C (λ_k)

$$C w_k = \lambda_k w_k \quad \text{(by definition)}$$

$$\left(\frac{1}{n}\sum_{i=1}^{n} x_i x_i^T\right) w_k = \lambda_k w_k$$

$$w_k = \sum_{i=1}^{n} \left(\frac{x_i^T w_k}{n \lambda_k}\right) x_i$$

w_k is a linear combination of data points.

$$w_k = X \alpha_k \text{, for some } \alpha_k \in \mathbb{R}^n$$

$$\begin{bmatrix} | & & | \\ x_1 & & x_n \\ | & & | \end{bmatrix} \begin{bmatrix} \alpha_{k1} \\ \vdots \\ \alpha_{kn} \end{bmatrix} = \sum_{i=1}^{n} \alpha_{ki} x_i$$

How to find α_k?

$$K \alpha_k = (n \lambda_k) \alpha_k$$

Date:
No:

We need α_k that satisfies the above equation.

We know,
$$w_k = X\alpha_k$$
$$w_k^T w_k = (X\alpha_k)^T (X\alpha_k)$$
$$w_k^T w_k = \alpha_k^T (X^T X)\alpha_k$$
$$\boxed{1 = \alpha_k^T K \alpha_k}$$

Also, satisfies the above equation.

$C = \frac{1}{n} X X^T$ Eigenvectors $= \{w_1, \ldots, w_e\}$

Eigenvalues $= \{\lambda_1 \geq \lambda_2 \geq \lambda_3 \ldots \geq \lambda_e\}$

$X X^T = nC$ Eigenvectors $= \{w_1, \ldots, w_e\}$

Eigenvalues $= \{n\lambda_1 \geq n\lambda_2 \geq \ldots \geq n\lambda_e\}$

$X^T X$ Eigenvalues $= \{n\lambda_1 \geq n\lambda_2 \geq \ldots \geq n\lambda_e\}$

Eigenvectors $= \{\beta_1, \ldots, \beta_e\}$

$$\boxed{K \beta_k = (n\lambda_k)\beta_k}$$

Let $\alpha_k = \frac{\beta_k}{\sqrt{n\lambda_k}}$

Date:
No:

Dealing with high dimensions

Input: $D = \{x_1, \ldots, x_n\}$ $x_i \in \mathbb{R}^d$ $[d >> n]$

Step 1: Compute $K = X^T X$, $K \in \mathbb{R}^{n \times n}$

Step 2: Compute eigendecomposition of K
eigenvectors $\{B_1, \ldots, B_k\}$ corresponding to eigenvalues $\{n\lambda_1, \ldots, n\lambda_2\}$
$\rightarrow O(n^3)$

Step 3: Set $\alpha_k = \frac{B_k}{\sqrt{n\lambda_k}}$ $\forall k = 1, \ldots, l$

Step 4: $W_k = X\alpha_k$ $\forall k$

ISSUE 2 : NON-LINEAR RELATIONSHIPS

Idea: Transform features from low dimension $\mathbb{R}^d$ to high dimension $\mathbb{R}^D$

$$\underset{\mathbb{R}^d}{x} \xrightarrow{\phi} \underset{\mathbb{R}^D}{\phi(x)}$$

Dimension of the transformed subspace

$$D = {}^{d+p}C_p$$

$d \rightarrow$ Dimension of the source subspace
$p \rightarrow$ polynomial relationship sought

Date:
No:

Kernel function

Any function $k: \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ which is a valid map.

$$k(x, x') = (x^T x' + 1)^p \rightarrow \text{polynomial kernel}$$

$$k(x, x') = \exp\left(\frac{-\|x - x'\|^2}{2\sigma^2}\right) \rightarrow \text{radial basis / Gaussian kernel}$$

Given a function, how can we say if it is a valid kernel?

<u>Method 1</u> : Exhibit a map ϕ explicitly

<u>Method 2</u> : Mercer's Theorem

$\downarrow$

$k: \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ is a valid kernel iff.

- k is symmetric i.e. $k(x, x') = k(x', x)$
- For any dataset $\{x_1, \ldots, x_n\}$, the matrix $K \in \mathbb{R}^{n \times n}$ where $K_{ij} = k(x_i, x_j)$ is positive semi-definite i.e. all eigenvalues of K are non-negative

Date:
No:

KERNEL PCA

Input: $\{x_1, \dots, x_n\}$ $x_i \in \mathbb{R}^d$; kernel $k : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$

Step 1: - Compute $K \in \mathbb{R}^{n \times n}$, where $K_{ij} = k(x_i, x_j)$ $\forall\, i, j$

- Centre the kernel

Step 2: Compute $\{\beta_1, \dots, \beta_\ell\}$ eigenvectors and $\{n\lambda_1 \geq \dots \geq n\lambda_\ell\}$ eigenvalues of K and normalize to get

$$\forall u \quad \alpha_u = \frac{\beta_u}{\sqrt{n\lambda_u}}$$

Step 3:

$$w_k = \phi(x)\,\alpha_k$$

↳ Defeats the purpose because it needs $\phi(x)$

- We cannot reconstruct the eigenvectors of the co-variance matrix.

- But, we can still compute the 'compressed' representation.

Modified Step 3: Compute

$$\sum_{j=1}^{n} \alpha_{\beta j} K_{ij} \quad \forall\, \beta$$

$$x_i \in \mathbb{R}^d \to \left[\sum_{j=1}^{n} \alpha_{1j} K_{ij}, \sum_{j=1}^{n} \alpha_{2j} \alpha_{ij}, \dots, \sum_{j=1}^{n} \alpha_{\ell j} k_{ij} \right]$$

Date:
No:

Centering the kernel

Given $K \in \mathbb{R}^{n \times n}$, $K_{ij} = k(x_i, x_j) \; \forall i,j$

Create a new kernel which is centred

$$K^c_{ij} = K_{ij} - \theta_i 1_j^T - 1_i \theta_j^T + \rho 1_i 1_j^T$$

, where $\theta_i = \frac{1}{n} \sum K_{ik} \; \forall i$

$$\rho = \frac{1}{n^2} \sum_{i,j} K_{ij}$$

- First apply kernel and then center data points

Date:
No:

WEEK 2 - EXAM POINTERS

vector $\alpha = [\alpha_1, \alpha_2, \ldots, \alpha_n]$

$$\alpha = \frac{\beta_k}{\sqrt{\lambda_k}} = \frac{\text{unit eigenvector}}{\sqrt{\text{eigenvalue of } K}}$$

$\lambda_k \Rightarrow$ kth largest eigenvalue
$\beta_k \Rightarrow$ corresponding eigenvector

Is K a valid kernel?

K is a valid kernel if

$$k(x_1, x_2) = \phi(x_1)^T \phi(x_2)$$

Shape of matrix K

$$\underset{n \times n}{K} = \underset{n \times d}{X^T} \underset{d \times n}{X}$$

Element ij of K matrix in polynomial kernel of degree 2

$$k(x_1, x_2) = (x_1^T x_2 + 1)^2$$